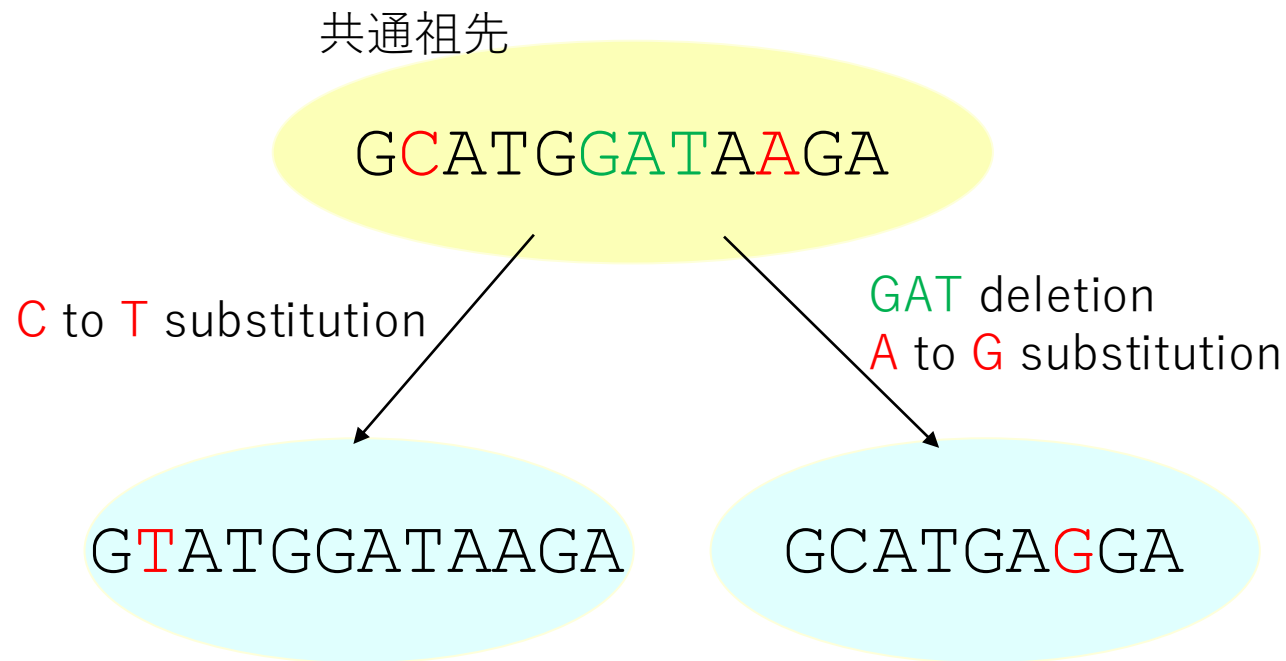
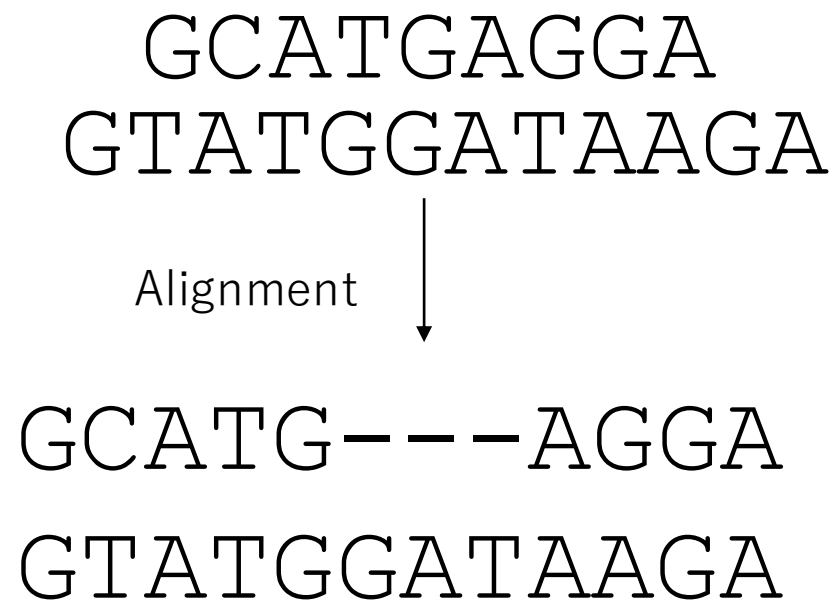


# バイオインフォマティクス 基本ツール (ホモロジー検索)

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# 配列比較とアライメント



相同性 (homology) : 共通祖先の同一の構造から派生した構造であること  
相同性は類似性 (similarity) に基づいて推定する

# 類似性スコアと配列アライメント

GCATGAGGA  
GTATGGATAAGA



Score definition : match **+2**、mismatch **-1**、gap **-2**

GCATG---AGGA

GTATGGATAAGA

**+2****-1****+2****+2****+2****-2****-2****-2****+2****-1****+2****+2** = **+6**

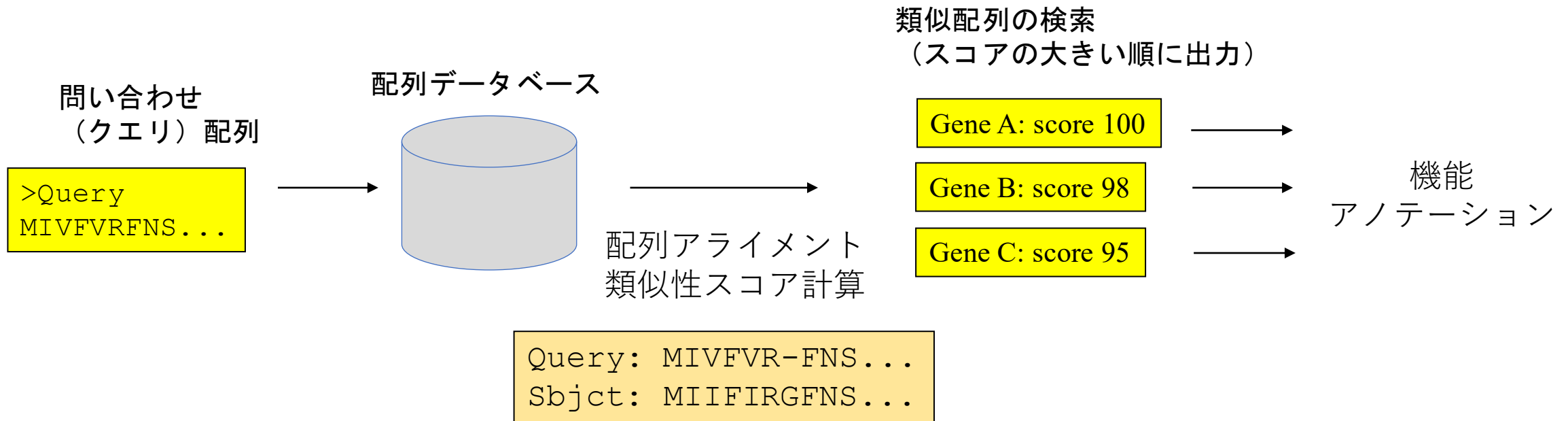
配列アライメント：類似性スコアが最大になるように各配列にギャップを挿入して並べる

# アミノ酸の類似性行列：BLOSUM62

|   |    |    |       |    |       |    |        |    |          |    |    |    |                   |    |    |       |    |   |          |                                                         |  |  |  |  |  |  |  |  |  |  |  |  |  |
|---|----|----|-------|----|-------|----|--------|----|----------|----|----|----|-------------------|----|----|-------|----|---|----------|---------------------------------------------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|
| P | 7  |    |       |    |       |    |        |    |          |    |    |    |                   |    |    |       |    |   |          | Higher score is assigned between<br>similar amino acids |  |  |  |  |  |  |  |  |  |  |  |  |  |
| A | -1 | 4  | small |    |       |    |        |    |          |    |    |    |                   |    |    |       |    |   |          |                                                         |  |  |  |  |  |  |  |  |  |  |  |  |  |
| G | -2 | 0  | 6     |    |       |    |        |    |          |    |    |    |                   |    |    |       |    |   |          |                                                         |  |  |  |  |  |  |  |  |  |  |  |  |  |
| N | -2 | -2 | 0     | 6  | amide |    |        |    |          |    |    |    |                   |    |    |       |    |   |          |                                                         |  |  |  |  |  |  |  |  |  |  |  |  |  |
| Q | -1 | -1 | -2    | 0  | 5     |    |        |    |          |    |    |    |                   |    |    |       |    |   |          |                                                         |  |  |  |  |  |  |  |  |  |  |  |  |  |
| D | -1 | -2 | -1    | 1  | 0     | 6  | acidic |    |          |    |    |    |                   |    |    |       |    |   |          |                                                         |  |  |  |  |  |  |  |  |  |  |  |  |  |
| E | -1 | -1 | -2    | 0  | 2     | 2  | 5      |    |          |    |    |    |                   |    |    |       |    |   |          |                                                         |  |  |  |  |  |  |  |  |  |  |  |  |  |
| T | -1 | 0  | -2    | 0  | -1    | -1 | -1     | 5  | hydroxyl |    |    |    |                   |    |    |       |    |   |          |                                                         |  |  |  |  |  |  |  |  |  |  |  |  |  |
| S | -1 | 1  | 0     | 1  | 0     | 0  | 0      | 1  | 4        |    |    |    |                   |    |    |       |    |   |          |                                                         |  |  |  |  |  |  |  |  |  |  |  |  |  |
| C | -3 | 0  | -3    | -3 | -3    | -3 | -4     | -1 | -1       | 9  |    |    |                   |    |    |       |    |   |          |                                                         |  |  |  |  |  |  |  |  |  |  |  |  |  |
| V | -2 | 0  | -3    | -3 | -2    | -3 | -2     | 0  | -2       | -1 | 4  |    |                   |    |    |       |    |   |          |                                                         |  |  |  |  |  |  |  |  |  |  |  |  |  |
| I | -3 | -1 | -4    | -3 | -3    | -3 | -3     | -1 | -2       | -1 | 3  | 4  | aliphatic (V,I,L) |    |    |       |    |   |          |                                                         |  |  |  |  |  |  |  |  |  |  |  |  |  |
| M | -2 | -1 | -3    | -2 | 0     | -3 | -2     | -1 | -1       | -1 | 1  | 1  | 5                 |    |    |       |    |   |          |                                                         |  |  |  |  |  |  |  |  |  |  |  |  |  |
| L | -3 | -1 | -4    | -3 | -2    | -4 | -3     | -1 | -2       | -1 | 1  | 2  | 2                 | 4  |    |       |    |   |          |                                                         |  |  |  |  |  |  |  |  |  |  |  |  |  |
| K | -1 | -1 | -2    | 0  | 1     | -1 | 1      | -1 | 0        | -3 | -2 | -3 | -1                | -2 | 5  | basic |    |   |          |                                                         |  |  |  |  |  |  |  |  |  |  |  |  |  |
| R | -2 | -1 | -2    | 0  | 1     | -2 | 0      | -1 | -1       | -3 | -3 | -3 | -1                | -2 | 2  | 5     |    |   |          |                                                         |  |  |  |  |  |  |  |  |  |  |  |  |  |
| H | -2 | -2 | -2    | 1  | 0     | -1 | 0      | -2 | -1       | -3 | -3 | -3 | -2                | -3 | -1 | 0     | 8  |   |          |                                                         |  |  |  |  |  |  |  |  |  |  |  |  |  |
| F | -4 | -2 | -3    | -3 | -3    | -3 | -3     | -2 | -2       | -2 | -1 | 0  | 0                 | 0  | -3 | -3    | -1 | 6 | aromatic |                                                         |  |  |  |  |  |  |  |  |  |  |  |  |  |
| Y | -3 | -2 | -3    | -2 | -1    | -3 | -2     | -2 | -2       | -2 | -1 | -1 | -1                | -1 | -2 | -2    | 2  | 3 | 7        |                                                         |  |  |  |  |  |  |  |  |  |  |  |  |  |
| W | -4 | -3 | -2    | -4 | -2    | -4 | -3     | -2 | -3       | -2 | -3 | -3 | -1                | -2 | -3 | -3    | -2 | 1 | 2        | 11                                                      |  |  |  |  |  |  |  |  |  |  |  |  |  |
|   | P  | A  | G     | N  | Q     | D  | E      | T  | S        | C  | V  | I  | M                 | L  | K  | R     | H  | F | Y        | W                                                       |  |  |  |  |  |  |  |  |  |  |  |  |  |

Higher score is assigned between similar amino acids

# 類似性（相同性）検索



# 問い合わせ配列の準備

## FASTA形式

```
>エントリー名 説明
配列.....
>エントリー名 説明
配列.....
.....
```

```
>gi|4758884|ref|NP_004553.1| parkin isoform 1 [Homo sapiens]
MIVFVRFNSSHGFPVEVDSDTSIFQLKEVVAKRQGV PADQLRVIFAGKELRNDWTVQNCDLDQQSIVHIV
QRPWRKGQEMNATGGDDPRNAAGGCEREPQSLTRVDLSSSVLPGDSVGLAVILHTDSRKDSPPAGSPAGR
SIYNSFYVYCKGPCQRVQPGKLRVQCSTCRQATLTLTQG PSCWDDVLI PNRM SGECQSPHCPGTSAEFFF
KCGAHPTSDKETPVALHLIATNSRNITCITCTDVRSPVLVFQCNSRHVICLDCFHLYCVTRLNDRQFVHD
PQLGYSLPCVAGCPNSLIKE LHHFRILGEEQYNRYQQYGAEECVLQMGGVLCPRPGCGAGLLPEPDQRKV
TCEGGNGLGCGFAFCRECKEAYHEGEC SAVFEASGTTTQAYRVDERAAEQARWEAASKETIKKTTKPCPR
CHVPVEKNGGCMHMKCPQPQCRLEWCWNCGCEWNRVCMGDHWF DV
```

# BLAST (Basic Local Alignment Search Tool)

The screenshot shows the NCBI BLAST homepage. At the top, there's a navigation bar with 'Home', 'Recent Results', 'Saved Strategies', and 'Help'. Below this, the 'Basic Local Alignment Search Tool' is introduced, explaining that BLAST finds regions of similarity between biological sequences. A 'Web BLAST' section offers three main options: 'Nucleotide BLAST' (nucleotide to nucleotide), 'blastx' (translated nucleotide to protein), and 'Protein BLAST' (protein to protein). There are also 'tblastn' and 'tblastx' options. A 'BLAST Genomes' section allows users to search by organism (Human, Mouse, Rat, Microbes). At the bottom, there's a 'Standalone and API BLAST' section with links to download BLAST, use the BLAST API, or use BLAST in the cloud. A 'Specialized searches' section lists various tools like SmartBLAST, Primer-BLAST, Global Align, CD-search, IgBLAST, VecScreen, CDART, and Multiple Alignment, each with a brief description of its function.

| program | query      | database   | translation |
|---------|------------|------------|-------------|
| BLASTP  | protein    | protein    |             |
| BLASTN  | nucleotide | nucleotide |             |
| BLASTX  | nucleotide | protein    | query       |
| TBLASTN | protein    | nucleotide | database    |
| TBLASTX | nucleotide | nucleotide | both        |

- 核酸配列は、相補鎖も同時に検索する
- 翻訳は、各鎖の3つの読み枠、計6通りの読み枠で比較する

<https://blast.ncbi.nlm.nih.gov/Blast.cgi>

# BLAST 検索インターフェイス

クエリ配列

BLAST<sup>®</sup> » blastp suite

blastn **blastp** blastx tblastn tblastx

**Enter Query Sequence**

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) [Clear](#)

```
>gi|4758884|ref|NP_004553.1| parkin isoform 1 [Homo sapiens]  
MIVFVRFNSSHGFPVEVSDTSIFQLKEVVAKRQGV PADQLRVIFAGKELRNDW  
TVQNCDL DQQSIVHIV  
QRPWRKGQEMNATGGDDPRNAAGGCEREPQSLTRVDLSSSVLP GDSVGLAVI  
LHTDSRKDSPAGSPAGR  
STNSP VYCKGPCQQRVQPGKLRVQCSTCRQATLTLTQGPSCWDDVLIPNRMS  
GECQSPHCPGTSAEFF  
KCGAHP TSDKETPVALHLIATNSRNITCITCTDVRSPVLVFCNSRHVICLDCFHL  
YCVTRLNDRQFVHD  
PQLGYSLPCVAGCPNSLIKELHHFRILGEEQYNRYQQYGAEECVLQMGGVLC  
RPGCGAGLLPEPDQRKV  
TCEGGNGLGCGFAFCRECKEAYHEGEC SAVFEASGTTTQAYRVD ERAAEQAR  
WEAASKETIKKTKPCPR  
CHVPVEKNGGCMHMKCPQPCRLEWCWNCGCEWNRVCMGDHWFVDV
```

Query subrange [?](#)

From

To

Or, upload file   [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#)

**Choose Search Set**

Database  [?](#)

Organism Optional

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown. [?](#)

**Program Selection**

Algorithm ☒ blastp (protein-protein BLAST)  
☐ PSI-BLAST (Position-Specific Iterated BLAST)  
Choose a BLAST algorithm [?](#)

**BLAST** Search database ClusteredNR using Blastp (protein-protein BLAST)  
☐ Show results in a new window

データベース



# 蛋白質配列データベース

- Non-redundant Protein (nr)
  - 公開されている蛋白質配列(DNA配列を翻訳したものを含む) すべてを統合して、重複 (完全に一致した配列) を除いたもの
- ClusteredNR (nr\_cluster\_seq) **[default]**
  - nr から、さらに90%以上一致した配列をまとめて重複を除いたもの
- RefSeq proteins (refseq\_protein)
  - 公的データベースに登録された配列データに対し、NCBIが生物種ごとにゲノムや遺伝子単位で再整理したデータベース (RefSeq) 中の蛋白質配列
- UniProtKB/Swiss-Prot (swissprot)
  - EBIで作成されている蛋白質配列データベース。専門家が手作業で注釈づけなどを行っているSwiss-Protと、DNA配列を機械的に翻訳、注釈づけして作成しているTrEMBLとからなる。

# BLAST 検索結果

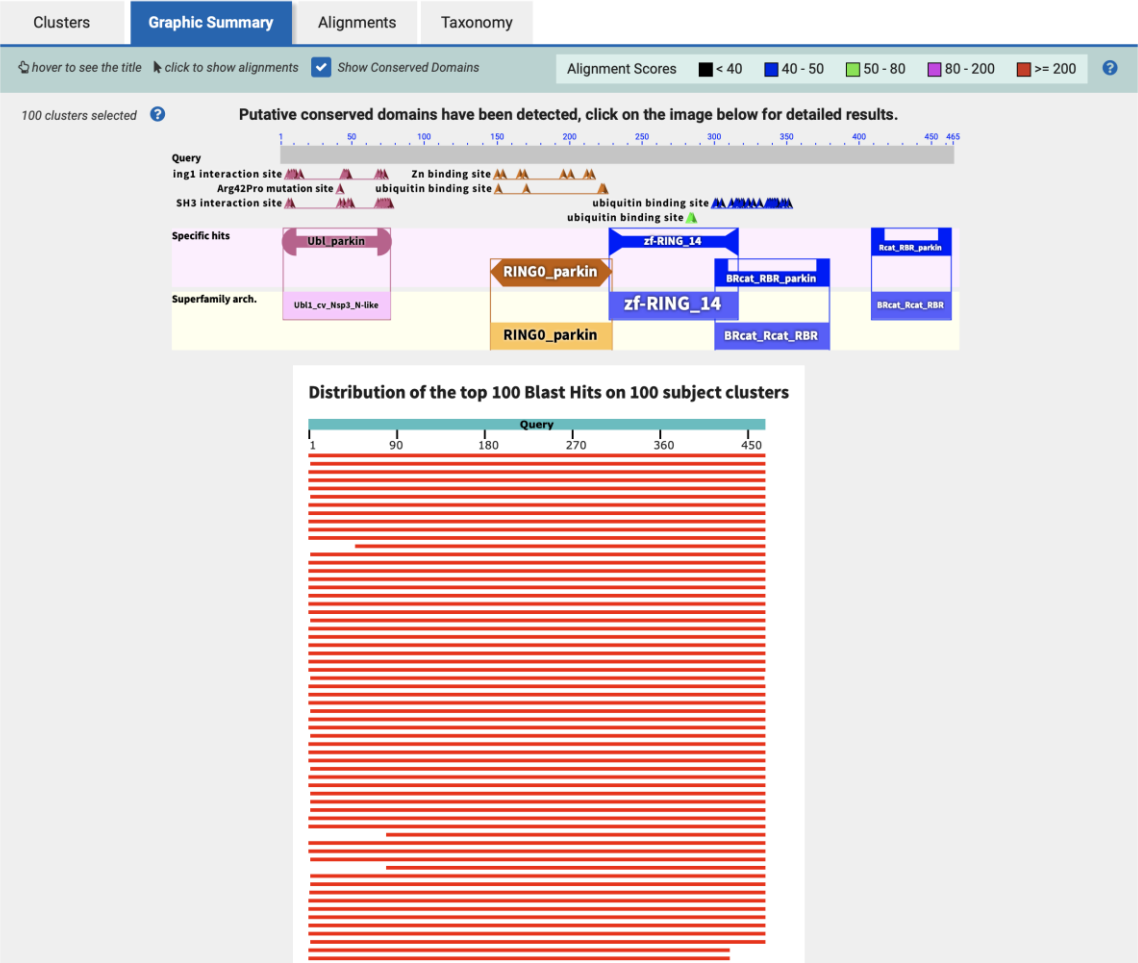
| Clusters                                                             |                                       | Graphic Summary             |                                                                       | Alignments |             | Taxonomy    |         |                |          |                          |  |                    |  |            |  |
|----------------------------------------------------------------------|---------------------------------------|-----------------------------|-----------------------------------------------------------------------|------------|-------------|-------------|---------|----------------|----------|--------------------------|--|--------------------|--|------------|--|
| Clusters producing significant alignments                            |                                       |                             |                                                                       |            |             | Download    |         | Select columns |          | Show                     |  | 100                |  |            |  |
| <input checked="" type="checkbox"/> select all 100 clusters selected |                                       |                             |                                                                       |            |             | GenPept     |         | Graphics       |          | Distance tree of results |  | Multiple alignment |  | MSA Viewer |  |
|                                                                      | Cluster Composition                   | Cluster Ancestor            | Cluster Representative Sequence                                       | Max Score  | Total Score | Query Cover | E value | Per. Ident     | Acc. Len | Accession                |  |                    |  |            |  |
|                                                                      | Click the to see the cluster contents |                             |                                                                       |            |             |             |         |                |          |                          |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 1 member(s), 1 organism(s)            | other sequences             | fusion protein HA-miRFP670nano-Parkin [synthetic construct]           | 975        | 975         | 100%        | 0.0     | 100.00%        | 627      | BET92570.1               |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 4 member(s), 4 organism(s)            | primates                    | E3 ubiquitin-protein ligase parkin isoform X10 [Macaca nemestr...     | 940        | 940         | 100%        | 0.0     | 97.19%         | 530      | XP_070952036.1           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 1 member(s), 1 organism(s)            | white-tufted-ear marmoset   | E3 ubiquitin-protein ligase parkin isoform X2 [Callithrix jacchus]    | 929        | 929         | 100%        | 0.0     | 95.70%         | 489      | XP_078226555.1           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 82 member(s), 26 organism(s)          | primates                    | E3 ubiquitin-protein ligase parkin isoform 2 [Homo sapiens]           | 894        | 894         | 100%        | 0.0     | 93.98%         | 437      | NP_054642.2              |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 8 member(s), 7 organism(s)            | prosimians                  | E3 ubiquitin-protein ligase parkin isoform 1 [Daubentonia mada...     | 870        | 870         | 100%        | 0.0     | 89.48%         | 465      | KAL2792878.1             |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 6 member(s), 1 organism(s)            | Philippine flying lemur     | E3 ubiquitin-protein ligase parkin isoform X4 [Cynocephalus vo...     | 858        | 858         | 100%        | 0.0     | 88.55%         | 480      | XP_062952603.1           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 30 member(s), 18 organism(s)          | even-toed ungulates & wh... | E3 ubiquitin-protein ligase parkin isoform X2 [Hippopotamus am...     | 854        | 854         | 100%        | 0.0     | 87.31%         | 495      | XP_057594137.1           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 12 member(s), 6 organism(s)           | odd-toed ungulates          | E3 ubiquitin-protein ligase parkin isoform X1 [Equus quagga]          | 853        | 853         | 100%        | 0.0     | 87.34%         | 466      | XP_046526636.1           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 2 member(s), 1 organism(s)            | Chinese tree shrew          | E3 ubiquitin-protein ligase parkin isoform X2 [Tupaia chinensis]      | 847        | 847         | 100%        | 0.0     | 87.74%         | 464      | XP_006152316.1           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 1 member(s), 1 organism(s)            | puma                        | E3 ubiquitin-protein ligase parkin [Puma concolor]                    | 843        | 843         | 100%        | 0.0     | 86.24%         | 691      | XP_025781231.1           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 1 member(s), 1 organism(s)            | gray mouse lemur            | E3 ubiquitin-protein ligase parkin isoform X2 [Microcebus murin...    | 842        | 842         | 100%        | 0.0     | 88.60%         | 464      | XP_075858924.1           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 9 member(s), 4 organism(s)            | primates                    | E3 ubiquitin-protein ligase parkin isoform X2 [Symphalangus sy...     | 839        | 839         | 90%         | 0.0     | 95.93%         | 442      | XP_055125023.2           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 48 member(s), 23 organism(s)          | carnivores                  | E3 ubiquitin-protein ligase parkin isoform X1 [Phoca vitulina]        | 838        | 838         | 100%        | 0.0     | 86.39%         | 469      | XP_032263409.1           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 2 member(s), 1 organism(s)            | gray squirrel               | E3 ubiquitin-protein ligase parkin isoform X2 [Sciurus carolinen...   | 838        | 838         | 100%        | 0.0     | 86.67%         | 462      | XP_047413015.1           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 1 member(s), 1 organism(s)            | American beaver             | E3 ubiquitin-protein ligase parkin isoform X1 [Castor canadensis]     | 836        | 836         | 100%        | 0.0     | 80.39%         | 509      | XP_073926901.1           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 5 member(s), 2 organism(s)            | elephants                   | E3 ubiquitin-protein ligase parkin isoform X1 [Elephas maximus...     | 833        | 833         | 100%        | 0.0     | 85.16%         | 465      | XP_049762059.1           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 26 member(s), 14 organism(s)          | cat family                  | E3 ubiquitin-protein ligase parkin isoform X1 [Felis catus]           | 830        | 830         | 100%        | 0.0     | 83.26%         | 481      | XP_044914560.1           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 5 member(s), 2 organism(s)            | pigs                        | E3 ubiquitin-protein ligase parkin [Sus scrofa]                       | 830        | 830         | 100%        | 0.0     | 86.02%         | 461      | NP_001038068.2           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 11 member(s), 4 organism(s)           | even-toed ungulates & wh... | E3 ubiquitin-protein ligase parkin isoform X2 [Camelus ferus]         | 828        | 828         | 100%        | 0.0     | 86.02%         | 460      | XP_032341198.1           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 2 member(s), 1 organism(s)            | prairie deer mouse          | E3 ubiquitin-protein ligase parkin isoform X2 [Peromyscus mani...     | 825        | 825         | 100%        | 0.0     | 82.24%         | 489      | XP_076408996.1           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 1 member(s), 1 organism(s)            | aardvark                    | E3 ubiquitin-protein ligase parkin [Orycteropus afer afer]            | 825        | 825         | 100%        | 0.0     | 84.23%         | 467      | XP_042637507.1           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 5 member(s), 1 organism(s)            | nine-banded armadillo       | E3 ubiquitin-protein ligase parkin isoform X5 [Dasypus novemci...     | 823        | 823         | 100%        | 0.0     | 84.30%         | 464      | XP_058145644.1           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 2 member(s), 2 organism(s)            | rabbits & hares             | E3 ubiquitin-protein ligase parkin isoform X1 [Ochotona princeps]     | 823        | 823         | 100%        | 0.0     | 84.95%         | 464      | XP_058512164.1           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 58 member(s), 22 organism(s)          | rodents                     | E3 ubiquitin-protein ligase parkin isoform 3 [Mus musculus]           | 820        | 820         | 100%        | 0.0     | 83.87%         | 465      | NP_001390399.1           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 4 member(s), 3 organism(s)            | dog, coyote, wolf, fox      | E3 ubiquitin-protein ligase parkin isoform X1 [Nyctereutes procy...   | 818        | 818         | 100%        | 0.0     | 81.95%         | 481      | XP_055168029.1           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 1 member(s), 1 organism(s)            | Arctic ground squirrel      | E3 ubiquitin-protein ligase parkin isoform 1-T1 [Urocitellus parryi]  | 818        | 818         | 100%        | 0.0     | 82.85%         | 482      | KAM4816522.1             |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 1 member(s), 1 organism(s)            | golden spiny mouse          | E3 ubiquitin-protein ligase parkin isoform X3 [Acomys russatus]       | 817        | 817         | 100%        | 0.0     | 83.44%         | 465      | XP_051020298.1           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 3 member(s), 1 organism(s)            | Nile rat                    | E3 ubiquitin-protein ligase parkin isoform X3 [Arvicanthis niloticus] | 815        | 815         | 99%         | 0.0     | 84.63%         | 461      | XP_076782759.1           |  |                    |  |            |  |
| <input checked="" type="checkbox"/>                                  | 9 member(s), 3 organism(s)            | bats                        | E3 ubiquitin-protein ligase parkin isoform X1 [Rhinolophus ferru...   | 814        | 814         | 100%        | 0.0     | 83.44%         | 463      | XP_032959339.1           |  |                    |  |            |  |

E-value

ランダムな配列で、その  
スコア以上の類似性を持つ  
個数の期待値

# BLAST 検索結果

## グラフィカル表示



## アライメント

Clusters Graphic Summary Alignments Taxonomy

Alignment view Pairwise Restore defaults Download

100 clusters selected

Download GenPept Graphics Next Previous Descriptions

**fusion protein HA-miRFP670nano-Parkin [synthetic construct]**

Sequence ID: [BET92570.1](#) Length: 627 Number of Matches: 1

Range 1: 163 to 627 GenPept Graphics Next Match Previous Match

|  | Score          | Expect | Method                       | Identities    | Positives     | Gaps      |
|--|----------------|--------|------------------------------|---------------|---------------|-----------|
|  | 975 bits(2520) | 0.0    | Compositional matrix adjust. | 465/465(100%) | 465/465(100%) | 0/465(0%) |

|       |     |                                                              |     |
|-------|-----|--------------------------------------------------------------|-----|
| Query | 1   | MIVFVRFNSSHGFPVEVSDTSIFQLKEVVAKRQGVADQLRVIFAGKELRNDWTVQNC    | 60  |
| Sbjct | 163 | MIVFVRFNSSHGFPVEVSDTSIFQLKEVVAKRQGVADQLRVIFAGKELRNDWTVQNC    | 222 |
| Query | 61  | LDQQSIVHIVORPWKQGEMNATGGDDPRNAAGGCEREPSLTRVDLSSSVLPDGSVGLA   | 120 |
| Sbjct | 223 | LDQQSIVHIVORPWKQGEMNATGGDDPRNAAGGCEREPSLTRVDLSSSVLPDGSVGLA   | 282 |
| Query | 121 | VILHTDSRKDSPAGSPAGRSIYNSFYVYCKGPCQVRQPGKLRVQCSTCRQATLTLTQGP  | 180 |
| Sbjct | 283 | VILHTDSRKDSPAGSPAGRSIYNSFYVYCKGPCQVRQPGKLRVQCSTCRQATLTLTQGP  | 342 |
| Query | 181 | SCWDDVLIIPNRMSGECQSPHCPGTSAEFFKCGAHPSTDKETSVALHLIATNSRNITCIT | 240 |
| Sbjct | 343 | SCWDDVLIIPNRMSGECQSPHCPGTSAEFFKCGAHPSTDKETSVALHLIATNSRNITCIT | 402 |
| Query | 241 | CTDVRSPVLVFQNSRHVICLDCFHLYCVTRLNDRQFVHDPQLGYSLPCVAGCPNSLIKE  | 300 |
| Sbjct | 403 | CTDVRSPVLVFQNSRHVICLDCFHLYCVTRLNDRQFVHDPQLGYSLPCVAGCPNSLIKE  | 462 |
| Query | 301 | LHHFRILGEEQYNYQQYGAECEVLQMGVLCPRPGCGAGLLPEPDQRKVTCEGNGLGC    | 360 |
| Sbjct | 463 | LHHFRILGEEQYNYQQYGAECEVLQMGVLCPRPGCGAGLLPEPDQRKVTCEGNGLGC    | 522 |
| Query | 361 | GFAFCRECKEAYHEGECSAVFEASGTTTQAYRVDERAAEQARWEAASKETIKKTKPCPR  | 420 |
| Sbjct | 523 | GFAFCRECKEAYHEGECSAVFEASGTTTQAYRVDERAAEQARWEAASKETIKKTKPCPR  | 582 |
| Query | 421 | CHVPVEKNGGCMHMKCPQPCRLEWCNCGCEWNRVCMGDHWFV                   | 465 |
| Sbjct | 583 | CHVPVEKNGGCMHMKCPQPCRLEWCNCGCEWNRVCMGDHWFV                   | 627 |

Download GenPept Graphics Next Previous Descriptions

**E3 ubiquitin-protein ligase parkin isoform X10 [Macaca nemestrina]**

Sequence ID: [XP\\_070952036.1](#) Length: 530 Number of Matches: 1

Range 1: 68 to 530 GenPept Graphics Next Match Previous Match

|  | Score          | Expect | Method                       | Identities   | Positives    | Gaps      |
|--|----------------|--------|------------------------------|--------------|--------------|-----------|
|  | 940 bits(2430) | 0.0    | Compositional matrix adjust. | 450/463(97%) | 453/463(97%) | 0/463(0%) |

**Related Information**

[Genome Data Viewer](#) - aligned genomic context

# グローバルアライメントとローカルアライメント

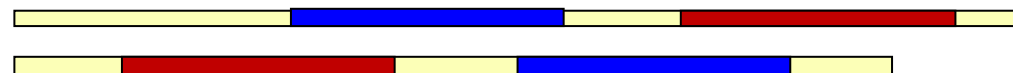
- グローバルアライメント
  - 配列全体を比較する



- ローカルアライメント
  - 類似性が最も高くなるような部分配列をとる



- ホモロジー検索では、ローカルアライメントを求める
  - アラインされる領域(HSP) が複数見つかることもある



# コマンドライン BLAST

プログラム名 -query クエリ配列ファイル -db データベース名

プログラム名： blastp, blastn, tblastn, blastx, tblastx

- SwissProt データベースに対して検索を実行する。  
CPUコア 4つを使って実行する (-num\_threads)。  

```
% blastp -query query.fa -db swissprot -num_threads 4 > blastout
```
- 同じ検索を行い、テーブル形式で出力する (-outfmt 6)。  

```
% blastp -query query.fa -db swissprot -num_threads 4 -outfmt 6 > blasttab
```

# BLASTの出力

|                                                                                                                            |                                                               |                 |            |  |
|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-----------------|------------|--|
| BLASTP 2.16.0+                                                                                                             |                                                               |                 |            |  |
| Database: SWISS-PROT: SWISS-PROT protein sequence database Release 2025_04<br>573,661 sequences; 207,922,125 total letters |                                                               |                 |            |  |
| Query= gi 4758884 ref NP_004553.1  parkin isoform 1 [Homo sapiens]                                                         |                                                               |                 |            |  |
| Length=465                                                                                                                 |                                                               |                 |            |  |
| Sequences producing significant alignments:                                                                                |                                                               | Score<br>(Bits) | E<br>Value |  |
| O60260                                                                                                                     | [PRKN_HUMAN] RecName: Full=E3 ubiquitin-protein ligase par... | 964             | 0.0        |  |
| Q9JK66                                                                                                                     | [PRKN_RAT] RecName: Full=E3 ubiquitin-protein ligase parki... | 830             | 0.0        |  |
| Q9WVS6                                                                                                                     | [PRKN_MOUSE] RecName: Full=E3 ubiquitin-protein ligase par... | 812             | 0.0        |  |
| Q7KTX7                                                                                                                     | [PRKN_DROME] RecName: Full=E3 ubiquitin-protein ligase par... | 382             | 7e-128     |  |
| E0VIU9                                                                                                                     | [PRKN_PEDHC] RecName: Full=E3 ubiquitin-protein ligase par... | 344             | 3e-113     |  |
| Q6T486                                                                                                                     | [RBRA_DICDI] RecName: Full=Probable E3 ubiquitin-protein l... | 77.8            | 5e-14      |  |
| Q84RR0                                                                                                                     | [ARI7_ARATH] RecName: Full=Probable E3 ubiquitin-protein l... | 77.0            | 9e-14      |  |
| Q6PFJ9                                                                                                                     | [ARI1_DANRE] RecName: Full=E3 ubiquitin-protein ligase ari... | 70.5            | 1e-11      |  |
| Q9Z1K5                                                                                                                     | [ARI1_MOUSE] RecName: Full=E3 ubiquitin-protein ligase ari... | 70.5            | 1e-11      |  |
| A2VEA3                                                                                                                     | [ARI1_BOVIN] RecName: Full=E3 ubiquitin-protein ligase ARI... | 70.5            | 1e-11      |  |
| Q9Y4X5                                                                                                                     | [ARI1_HUMAN] RecName: Full=E3 ubiquitin-protein ligase ARI... | 70.5            | 1e-11      |  |
| Q94981                                                                                                                     | [ARI1_DROME] RecName: Full=E3 ubiquitin-protein ligase ari... | 70.1            | 1e-11      |  |
| Q6NW85                                                                                                                     | [ARI1L_DANRE] RecName: Full=E3 ubiquitin-protein ligase ar... | 70.1            | 1e-11      |  |
| B1H1E4                                                                                                                     | [ARI1_XENTR] RecName: Full=E3 ubiquitin-protein ligase ari... | 69.7            | 2e-11      |  |
| Q32NS4                                                                                                                     | [ARI1_XENLA] RecName: Full=E3 ubiquitin-protein ligase ari... | 68.6            | 4e-11      |  |
| Q8L829                                                                                                                     | [ARI5_ARATH] RecName: Full=Probable E3 ubiquitin-protein l... | 67.0            | 1e-10      |  |
| Q9UBS8                                                                                                                     | [RNF14_HUMAN] RecName: Full=E3 ubiquitin-protein ligase RN... | 65.5            | 3e-10      |  |
| Q9Z1K6                                                                                                                     | [ARI2_MOUSE] RecName: Full=E3 ubiquitin-protein ligase ARI... | 65.1            | 5e-10      |  |
| POC8K8                                                                                                                     | [ARI6_ARATH] RecName: Full=Putative E3 ubiquitin-protein l... | 64.7            | 8e-10      |  |
| O95376                                                                                                                     | [ARI2_HUMAN] RecName: Full=E3 ubiquitin-protein ligase ARI... | 64.7            | 8e-10      |  |
| Q9SKC3                                                                                                                     | [ARI9_ARATH] RecName: Full=Probable E3 ubiquitin-protein l... | 63.5            | 2e-09      |  |
| Q9JI90                                                                                                                     | [RNF14_MOUSE] RecName: Full=E3 ubiquitin-protein ligase RN... | 61.6            | 6e-09      |  |
| F1RB95                                                                                                                     | [RNF14_DANRE] RecName: Full=E3 ubiquitin-protein ligase RN... | 61.2            | 9e-09      |  |
| Q05120                                                                                                                     | [UBIL_NPPOP] RecName: Full=Ubiquitin-like protein; Flags: ... | 55.5            | 1e-08      |  |

|                                                                                                                                                                                                                                                               |     |                                                                 |     |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----------------------------------------------------------------|-----|--|
| >Q7KTX7 [PRKN_DROME] RecName: Full=E3 ubiquitin-protein ligase<br>parkin {ECO:0000303 PubMed:12642658}; EC=2.3.2.31 {ECO:0000269 PubMed:17456438,<br>ECO:0000269 PubMed:20194754, ECO:0000269 PubMed:23770917,<br>ECO:0000269 PubMed:24901221};<br>Length=482 |     |                                                                 |     |  |
| Score = 382 bits (981), Expect = 7e-128, Method: Compositional matrix adjust.<br>Identities = 200/474 (42%), Positives = 280/474 (59%), Gaps = 33/474 (7%)                                                                                                    |     |                                                                 |     |  |
| Query                                                                                                                                                                                                                                                         | 1   | MIVFVRFNSSHGFPVEVSDSTSIFQLKEVVAKRQGVPADQLRVIFAGKELRNDWTVQNCD    | 60  |  |
|                                                                                                                                                                                                                                                               |     | + ++V+ N+ V ++ I +KE+VA + G+ D L++IFAGKEL + T++ CD              |     |  |
| Sbjct                                                                                                                                                                                                                                                         | 30  | LSIYVKTNNTGKTLTVNLEPQWDIKNVKELVAPQLGLQPDLDKIIFAGKELSDATTIEQCD   | 89  |  |
|                                                                                                                                                                                                                                                               |     |                                                                 |     |  |
| Query                                                                                                                                                                                                                                                         | 61  | LDQQSIVHIVQ-RPWRKGQEMNATGGDDPRNAAGGCERE PQSLTRVDLSSSVLPGDSVGL   | 119 |  |
|                                                                                                                                                                                                                                                               |     | L QQS++H ++ RP + Q++ + ++ + +P + T +DL                          |     |  |
| Sbjct                                                                                                                                                                                                                                                         | 90  | LQQQSVLHAIRLRPPVQRQKIQSATLEEEEP SLSDEASKPLNETLLDLQ-----         | 138 |  |
|                                                                                                                                                                                                                                                               |     |                                                                 |     |  |
| Query                                                                                                                                                                                                                                                         | 120 | AVILHTDSRKDSP PAGSPAGRSIYNSFYVYCKGPCQRVQPGKLRVQCSTCRQATLTLTQG   | 179 |  |
|                                                                                                                                                                                                                                                               |     | L ++ R + ++ F+V+C C ++ GKLRV+C+ C+ T+ +                         |     |  |
| Sbjct                                                                                                                                                                                                                                                         | 139 | ---LESEERLNITDEERVRAKA---HFFVHC-SQCDKLCNGKLRVRCALCKGGAFTVHRD    | 191 |  |
|                                                                                                                                                                                                                                                               |     |                                                                 |     |  |
| Query                                                                                                                                                                                                                                                         | 180 | PSCWDDVLIPNRMSGECQSPHC PGTS-----AEFFFKCGAHPTS-DKETPVALHLIA      | 230 |  |
|                                                                                                                                                                                                                                                               |     | P CWDDVL R+ G C+S AEFFFKC H + +K+ L+LI                          |     |  |
| Sbjct                                                                                                                                                                                                                                                         | 192 | PECWDDVLKSRRI PGHCESLEVACVDNAAGDPPFAEFFFFKCAEHVSGGEKDFAAPLNLIK  | 251 |  |
|                                                                                                                                                                                                                                                               |     |                                                                 |     |  |
| Query                                                                                                                                                                                                                                                         | 231 | TNSRNITCITCTDVRSPVLVFQCN SRHVICLDCFHLYCVTRLNDRQFVHDPQLGYS LPCV  | 290 |  |
|                                                                                                                                                                                                                                                               |     | N +N+ C+ CTDV VLVF C S+HV C+DCF YC +RL +RQF+ P GY+LPC           |     |  |
| Sbjct                                                                                                                                                                                                                                                         | 252 | NNIKNVPC LACTDVS DTVLVFPCASQHVTCIDCFRHYCRSRLGERQFMPHPDFGYTLPCP  | 311 |  |
|                                                                                                                                                                                                                                                               |     |                                                                 |     |  |
| Query                                                                                                                                                                                                                                                         | 291 | AGCPNSLIKELH HFRILGEEQYNRYQQYGAE ECVLMGGVLCPRPGCGAGLLPEPDQRKV   | 350 |  |
|                                                                                                                                                                                                                                                               |     | AGC +S I+E+HHF++L E+Y+RYQ++ EE VLQ GGVLCF+PGCG GLL EPD R KV     |     |  |
| Sbjct                                                                                                                                                                                                                                                         | 312 | AGCEHSFIEEIH FKLTLTREEY DRYQRFATEEYVLQAGGVLC PQPGCGMGLLVEPDCKRV | 371 |  |
|                                                                                                                                                                                                                                                               |     |                                                                 |     |  |
| Query                                                                                                                                                                                                                                                         | 351 | TCEGGNGLGCGFAFCRECKEAYHEGECSAVFE-ASGTTTQAYRVDERAAEQARWEAASKE    | 409 |  |
|                                                                                                                                                                                                                                                               |     | TC+ GCG+ FCR C + YH GEC AS T + Y VD A +ARW+ AS                  |     |  |
| Sbjct                                                                                                                                                                                                                                                         | 372 | TCQN----GCGYVFCRNCLQGYHIGEC LPEGTGASATNSCEYTVDPNRAAEARWDEASNV   | 427 |  |
|                                                                                                                                                                                                                                                               |     |                                                                 |     |  |
| Query                                                                                                                                                                                                                                                         | 410 | TIKKTTKPCPRCHVPVEKNGGCMHMKCPQPQCRLEWCWNCGCEWNRVCMGDHWF          | 463 |  |
|                                                                                                                                                                                                                                                               |     | TIK +TKPCP+C P E++GGCMHM C + C EWCW C EW R CMG HWF              |     |  |
| Sbjct                                                                                                                                                                                                                                                         | 428 | TIKVSTKPCPKCRTPTERDGGCMH MVCTRAGCGFEWCWVCQTEWTRDCMGAHWF         | 481 |  |

# テーブル形式出力 (BLAST Tabular format)

| 1. query                   |  | 2. subject<br>(database) | 3. %identity | 4. align-len | 5. mismatch | 6. gap_open | 7. q_start | 8. q_end | 9. s_start | 10. s_end | 11. eval  | 12. bit-score |
|----------------------------|--|--------------------------|--------------|--------------|-------------|-------------|------------|----------|------------|-----------|-----------|---------------|
| gi 4758884 ref NP_004553.1 |  | O60260                   | 99.785       | 465          | 1           | 0           | 1          | 465      | 1          | 465       | 0.0       | 964           |
| gi 4758884 ref NP_004553.1 |  | Q9JK66                   | 84.946       | 465          | 70          | 0           | 1          | 465      | 1          | 465       | 0.0       | 830           |
| gi 4758884 ref NP_004553.1 |  | Q9WVS6                   | 83.441       | 465          | 76          | 1           | 1          | 465      | 1          | 464       | 0.0       | 812           |
| gi 4758884 ref NP_004553.1 |  | Q7KTX7                   | 42.194       | 474          | 241         | 8           | 1          | 463      | 30         | 481       | 7.05e-128 | 382           |
| gi 4758884 ref NP_004553.1 |  | E0VIU9                   | 38.445       | 463          | 249         | 9           | 3          | 463      | 32         | 460       | 2.53e-113 | 344           |
| gi 4758884 ref NP_004553.1 |  | Q6T486                   | 27.823       | 248          | 139         | 10          | 219        | 453      | 117        | 337       | 5.20e-14  | 77.8          |
| gi 4758884 ref NP_004553.1 |  | Q84RR0                   | 29.004       | 231          | 125         | 10          | 231        | 453      | 130        | 329       | 9.49e-14  | 77.0          |
| gi 4758884 ref NP_004553.1 |  | Q6PFJ9                   | 26.500       | 200          | 119         | 7           | 257        | 453      | 175        | 349       | 9.81e-12  | 70.5          |
| gi 4758884 ref NP_004553.1 |  | Q9Z1K5                   | 26.500       | 200          | 119         | 7           | 257        | 453      | 203        | 377       | 1.01e-11  | 70.5          |
| gi 4758884 ref NP_004553.1 |  | A2VEA3                   | 26.500       | 200          | 119         | 7           | 257        | 453      | 203        | 377       | 1.01e-11  | 70.5          |
| gi 4758884 ref NP_004553.1 |  | Q9Y4X5                   | 26.500       | 200          | 119         | 7           | 257        | 453      | 205        | 379       | 1.07e-11  | 70.5          |
| gi 4758884 ref NP_004553.1 |  | Q94981                   | 26.538       | 260          | 158         | 11          | 209        | 464      | 105        | 335       | 1.34e-11  | 70.1          |
| gi 4758884 ref NP_004553.1 |  | Q6NW85                   | 26.500       | 200          | 119         | 7           | 257        | 453      | 181        | 355       | 1.35e-11  | 70.1          |
| gi 4758884 ref NP_004553.1 |  | B1H1E4                   | 26.500       | 200          | 119         | 7           | 257        | 453      | 177        | 351       | 1.75e-11  | 69.7          |
| gi 4758884 ref NP_004553.1 |  | Q32NS4                   | 26.066       | 211          | 126         | 8           | 257        | 464      | 177        | 360       | 4.50e-11  | 68.6          |
| gi 4758884 ref NP_004553.1 |  | Q8L829                   | 27.826       | 230          | 127         | 10          | 232        | 453      | 126        | 324       | 1.33e-10  | 67.0          |
| gi 4758884 ref NP_004553.1 |  | Q9UBS8                   | 28.319       | 226          | 129         | 8           | 256        | 464      | 239        | 448       | 3.33e-10  | 65.5          |
| gi 4758884 ref NP_004553.1 |  | Q9Z1K6                   | 23.137       | 255          | 157         | 9           | 203        | 453      | 110        | 329       | 4.63e-10  | 65.1          |
| gi 4758884 ref NP_004553.1 |  | P0C8K8                   | 28.017       | 232          | 129         | 11          | 230        | 453      | 125        | 326       | 7.87e-10  | 64.7          |
| gi 4758884 ref NP_004553.1 |  | O95376                   | 24.016       | 254          | 156         | 9           | 203        | 453      | 111        | 330       | 7.95e-10  | 64.7          |
| gi 4758884 ref NP_004553.1 |  | Q9SKC3                   | 26.786       | 224          | 124         | 10          | 235        | 449      | 125        | 317       | 1.90e-09  | 63.5          |
| gi 4758884 ref NP_004553.1 |  | Q9JI90                   | 26.009       | 223          | 136         | 7           | 261        | 464      | 237        | 449       | 6.37e-09  | 61.6          |
| gi 4758884 ref NP_004553.1 |  | F1RB95                   | 26.744       | 172          | 104         | 5           | 306        | 462      | 266        | 430       | 8.54e-09  | 61.2          |
| gi 4758884 ref NP_004553.1 |  | Q05120                   | 33.333       | 72           | 48          | 0           | 1          | 72       | 1          | 72        | 1.27e-08  | 55.5          |

この出力形式は柔軟に変更できる。blastp -help を参照。

例えば、オプションを -outfmt "6 std stitle" とすると、上記の出力に加えて、13カラム目にデータベース配列のタイトル行を付加することができる。

# コマンドライン BLAST

## データベースを自前で用意する

```
makeblastdb -in 配列ファイル -dbtype {prot|nucl} -out データベース名
```

- 配列ファイル `sce.fa`（出芽酵母 *Saccharomyces cerevisiae* 全遺伝子の翻訳配列）にインデックスをつけ、`sce`というデータベースを作る。

```
% makeblastdb -in sce.fa -dbtype prot -parse_seqids -out sce
```

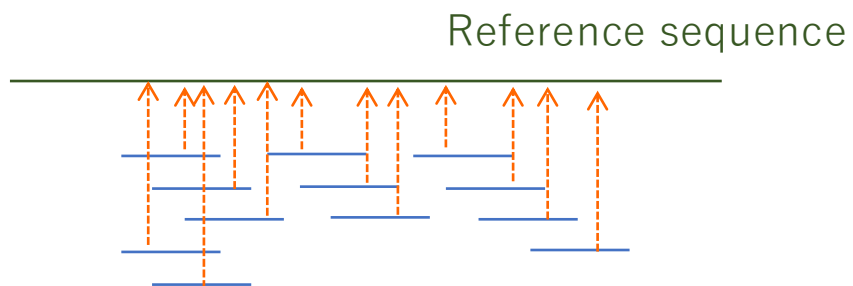
- 作成したデータベースに対して、`spo0.fa`（分裂酵母 *Schizosaccharomyces pombe* 遺伝子の一部）をクエリとして検索を実行する。E-valueの閾値を0.001とし(`-evalue`)、クエリ当たり上位10個までのヒットを出力(`-max_target_seqs`)。

```
% blastp -query spo0.fa -db sce -num_threads 4  
-outfmt 6 -evalue 0.001 -max_target_seqs 10 > spo0_sce.out
```



# 配列アライメントに基づく方法

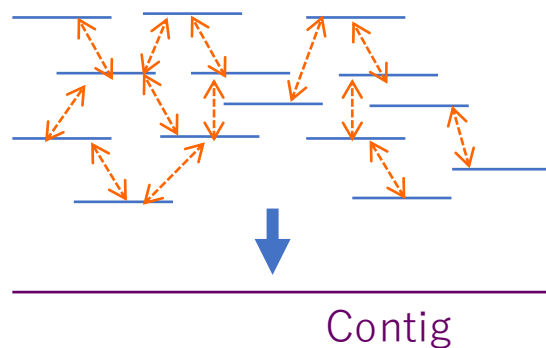
## リファレンスゲノムへのマッピング



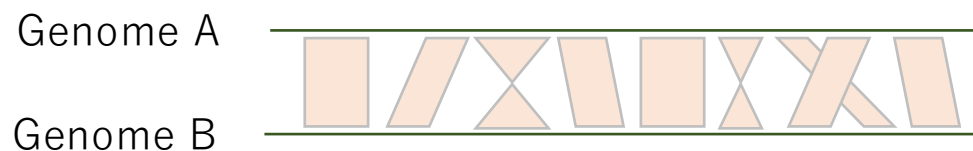
## マルチプルアライメント

|        |                                                                 |
|--------|-----------------------------------------------------------------|
| HBHU   | GDLSTPDAVMG--NPKVKAHG---KKVLGAFSDGLAHL---DNLKGTFFATLSELHCDKLH   |
| HEGT2  | GNLCSESAIMG--NPKVKAHG---RKVLNSFGNAIKHM---DDLKGTFFADLSELHCDKLH   |
| HAHU   | -----DL SHG--SAQVKGHG---KKVADALTNAVAHV---DDMPNALSALS DLHAHKLR   |
| HACH2  | -----DL SHG--SAQIKGHG---KKVVAALIEAANHI---DDIAGTLSKLS DLHAHKLR   |
| MYHU   | ---KSEDEMKA--SEDLKKHG---ATVLTALGGILKKK---GHHEAEIKPLAQSHATKHK    |
| GGNKT  | --ARLGDVSAGKDNSKLRGHSITL MYALQNF IDALDNV---DRLKCVVEKFAVNH INRQ- |
| S10238 | SFLKDFDEV PQ--NNPSLQAHA---EKVFGLVRDSAAQLRATGVVVLADASLGSVHVQKG-  |

## 配列アセンブリによる新規配列決定



## ゲノム配列比較



## モチーフ抽出 モチーフ検索

|   |            |
|---|------------|
| A | 0050000000 |
| R | 0000000000 |
| N | 0000000000 |
| D | 0000000000 |
| C | 0000050050 |
| : | :          |
| T | 0001000000 |
| W | 0200500000 |
| Y | 0100000000 |

## 系統樹作成

